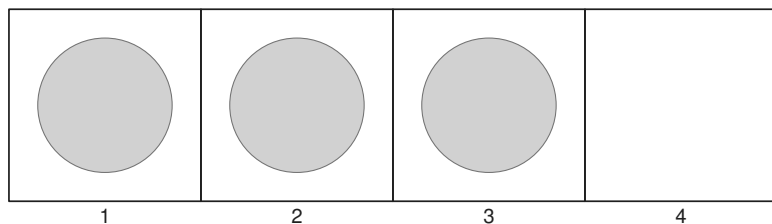


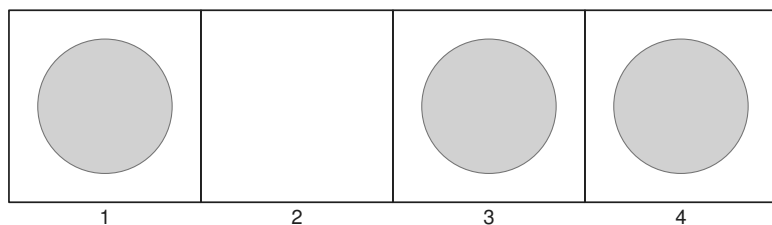
Problem 1: A penny may jump over the penny next to it, to the left or to the right, into the empty square right behind it. The penny that was jumped over comes off the strip. Put a penny on each circle of strip A. The last square is empty. Jump until no jump is left. Make one tally mark for each jump. Draw the pennies that are left as dots on the small strip and write how many in the number box. Then do the same on strip B, where the second square is empty.

A

jumps

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left

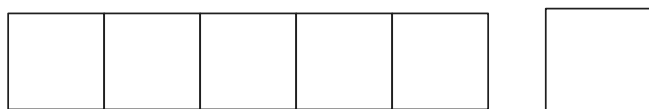
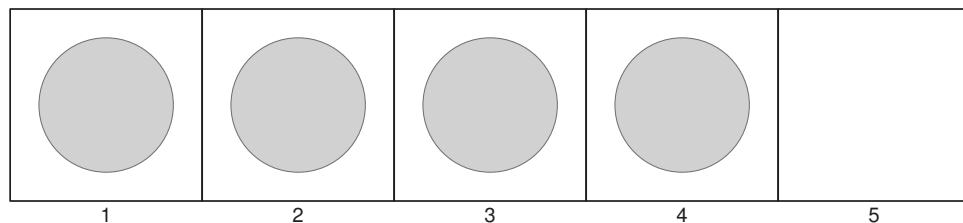
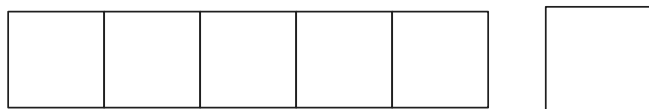
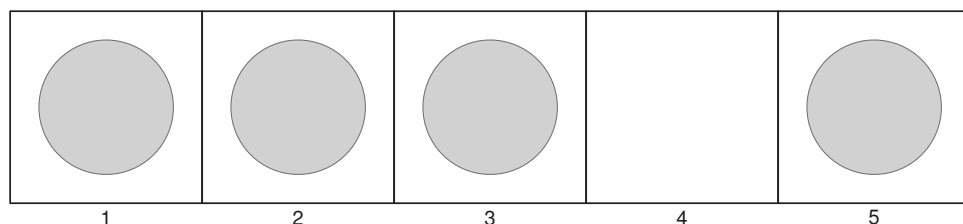
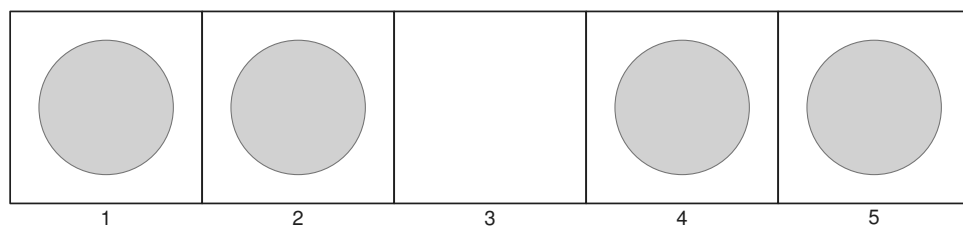
B

jumps

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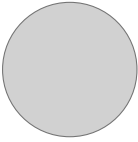
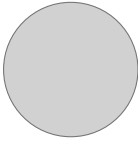
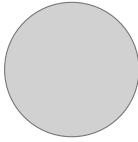
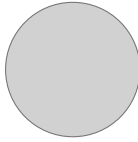
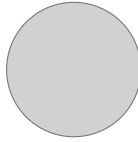
left

Problem 2: Strips C, D and E have five squares each, with the empty square in a different place on each. On each strip, put a penny on each circle and jump until no jump is left, trying to leave as few pennies as you can. Draw each ending on a small strip and write the number of pennies left. On strip E, play twice, starting with a different jump each time, and draw both endings. Then look at all your endings from Problems 1 and 2 side by side, and circle the letter of each strip that let you get down to one penny.

C**D****E**

Problem 3: Strips F, G and H have six squares each. On each strip, put a penny on each circle and play until no jump is left, trying to leave as few pennies as you can. Play each strip at least three times, making different choices where you can. Draw every different ending you find on the small strips under it and write its number of pennies. On strip F, circle the square where the last penny sat in each ending. On strip G, write beside each ending the number of the square its first jump started from. Write the fewest pennies you could leave on each strip in the number box at the right.

F

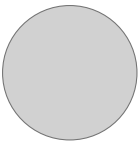
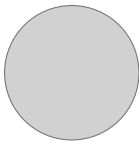
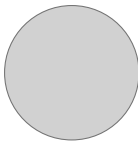
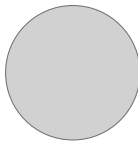
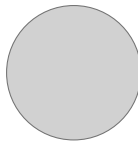
					
1	2	3	4	5	6

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G

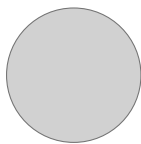
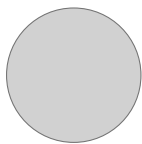
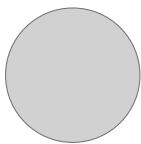
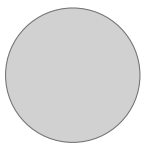
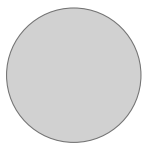
					
1	2	3	4	5	6

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H

					
1	2	3	4	5	6



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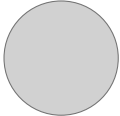
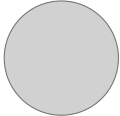
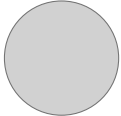
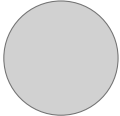
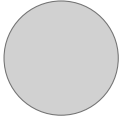
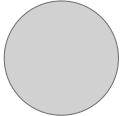
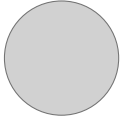
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Problem 4: Strip J has eight squares and the fourth square is empty. Put a penny on each circle and play three times, making different choices each time. Each time, make a tally mark for every jump, and when no jump is left, count the pennies. Fill in one row of the table for each play: the number of jumps, the number of pennies left, and the two numbers added together. Draw each ending on a small strip.

J

							
1	2	3	4	5	6	7	8

jumps	pennies left	jumps + pennies left

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jumps

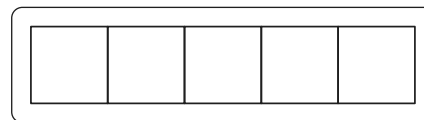
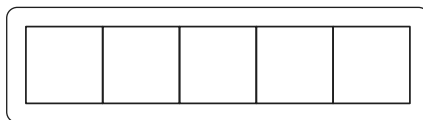
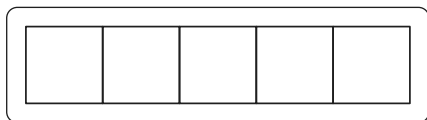
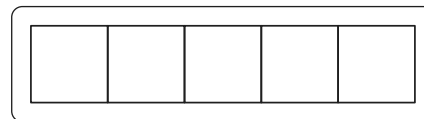
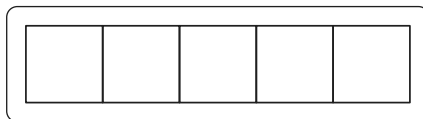
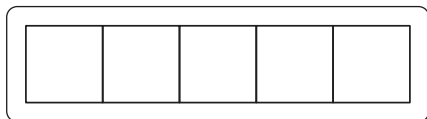
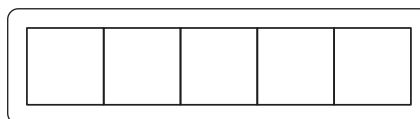
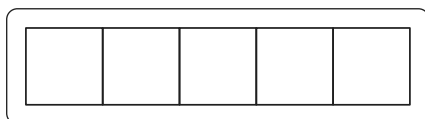
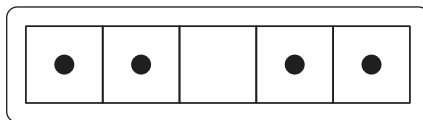
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jumps

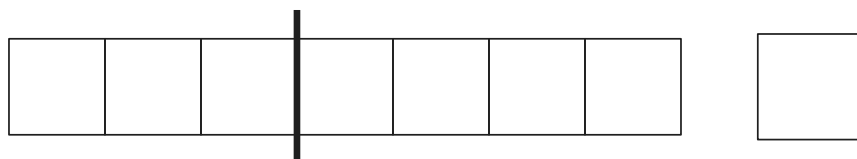
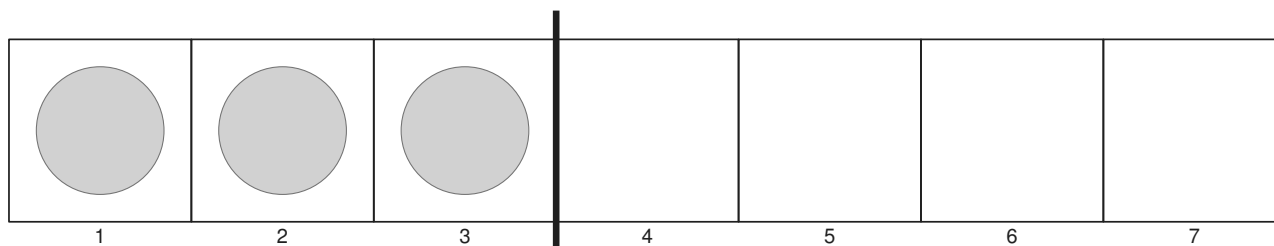
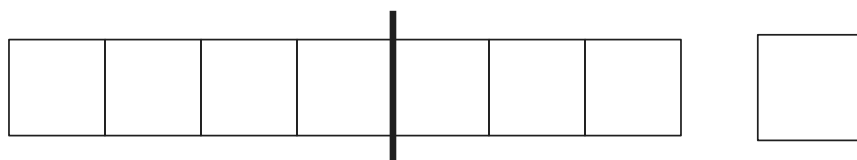
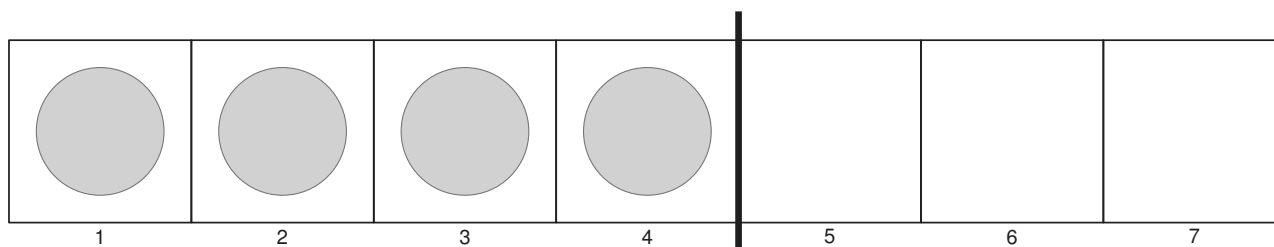
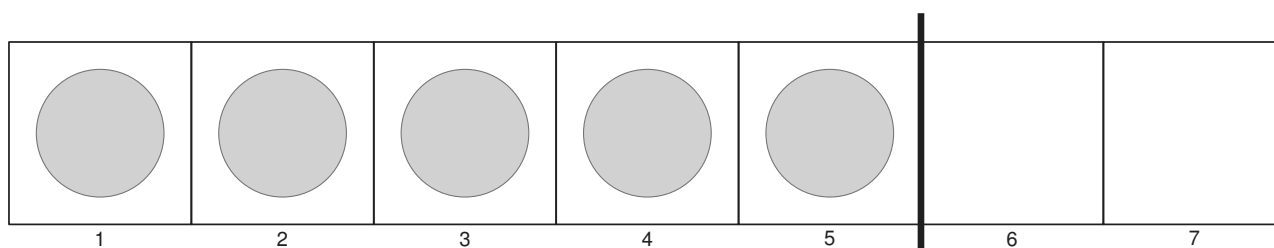
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jumps

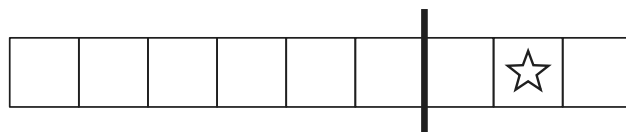
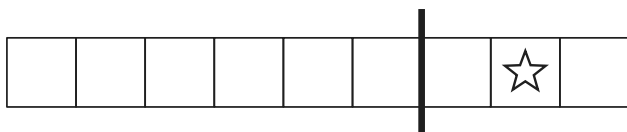
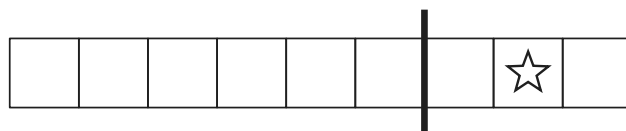
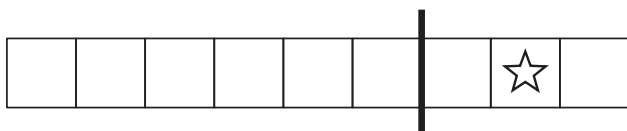
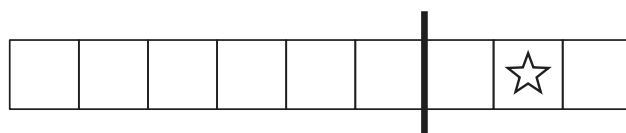
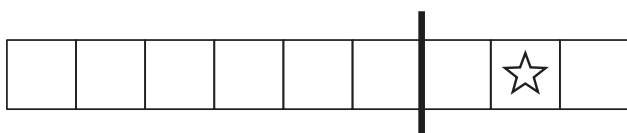
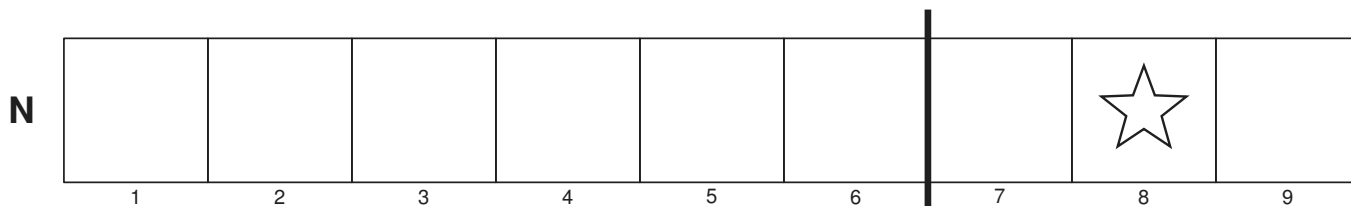
Problem 5: Make a map of every position you can reach on strip E, the strip of five with the middle square empty. The top box shows the start. In the boxes below, draw every position that one jump can reach from the start, with an arrow from the start to each one. Then, from each new position, draw every position that one more jump can reach, with an arrow. Keep going until every position is either already on the map or has no jump left. Circle the positions that have no jump left and write the number of pennies in each. Count the positions on your map and write the number in the number box.



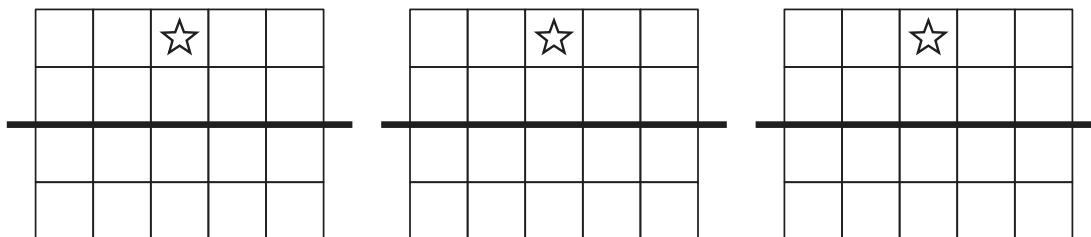
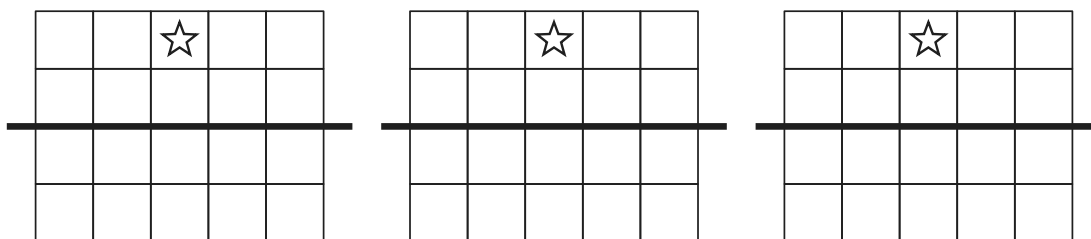
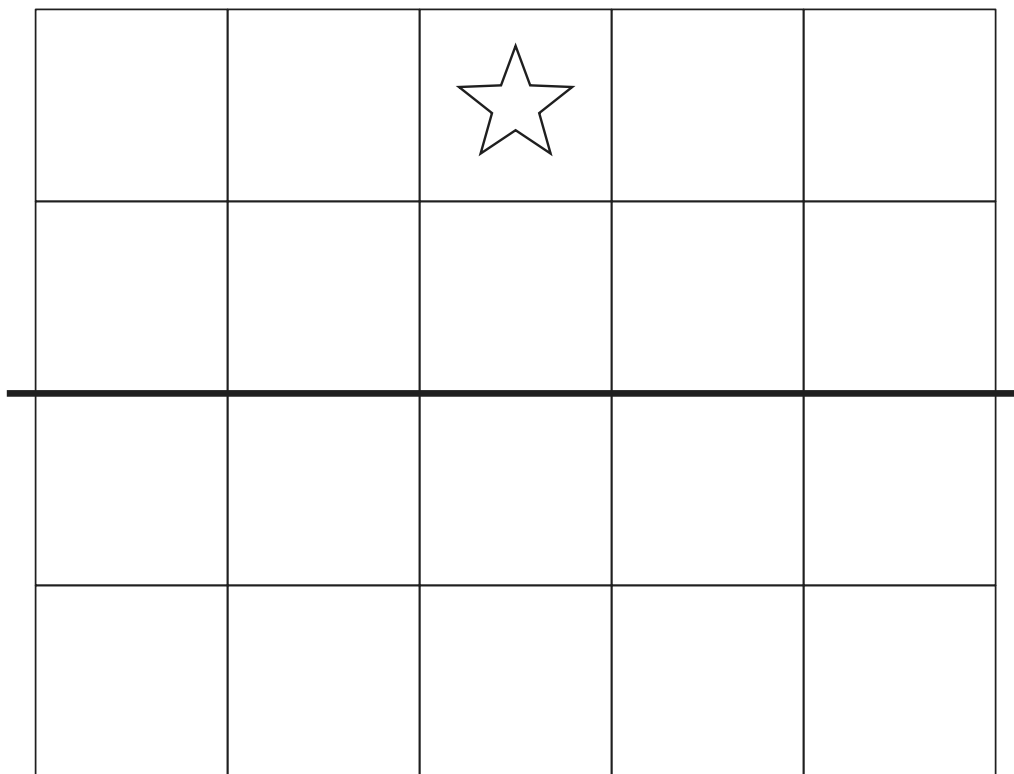
Problem 6: Strips K, L and M have a thick line across them. The pennies start on the left of the line, on the circles. Jump so that a penny gets as far to the right as it can. Play each strip twice. On the small strip under each, put a cross on the farthest square that any penny reached, and write in the number box how many squares past the line it is.

K**L****M**

Problem 7: Strip N has nine squares, a thick line after square 6, and a star on square 8. Put pennies on any of the squares 1 to 6, as many or as few as you like, and try to get a penny onto the star. For each army you try, draw its pennies as dots on one of the small strips and put a cross on the farthest square a penny reached. Try at least six different armies, some with gaps between the pennies.

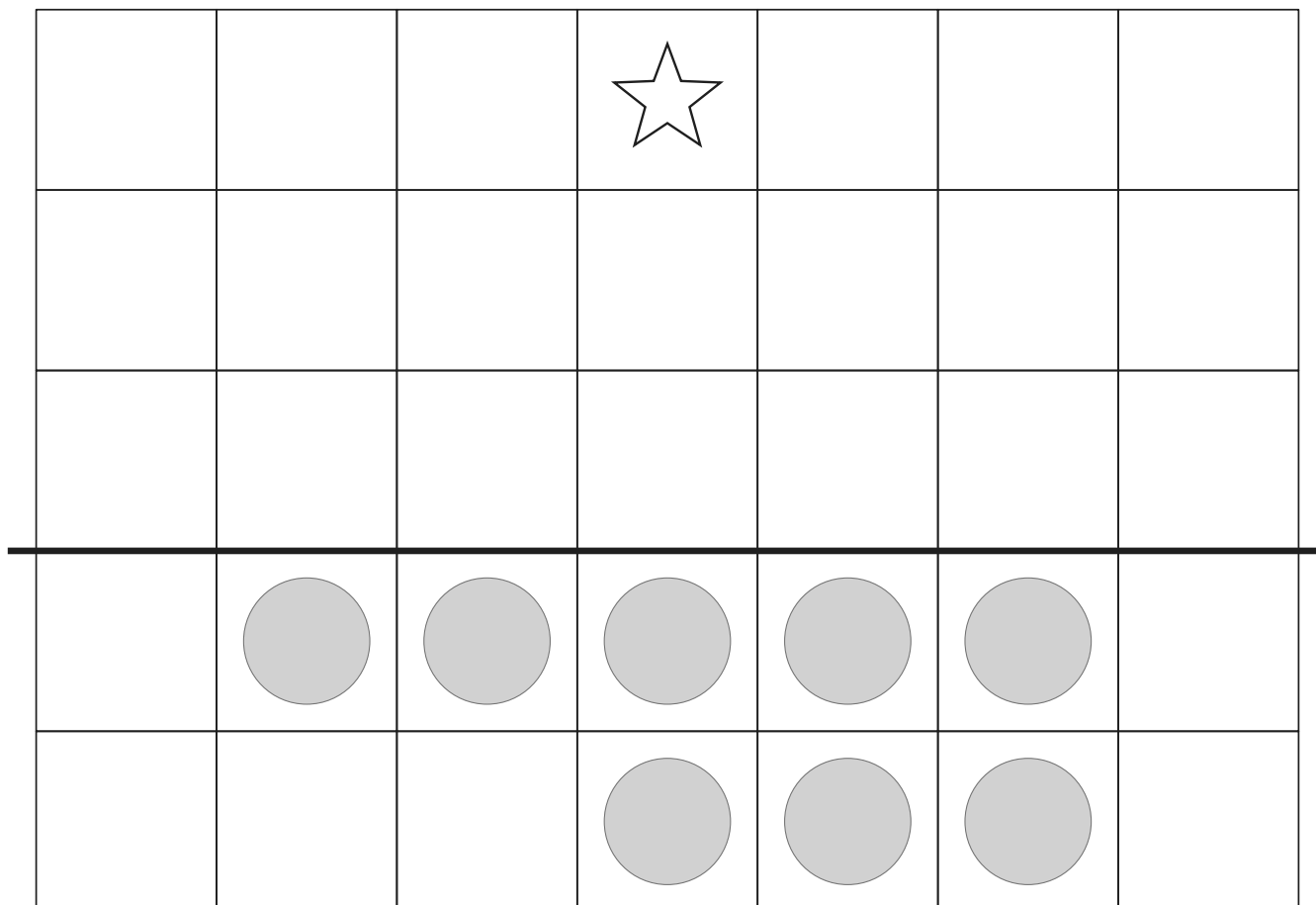


Problem 8: On board P the star is two rows above the line. Pennies may also jump up or down, over the penny next to them into the empty square right behind it. Take exactly three pennies. Put them anywhere below the line and try to land one on the star. Draw each army of three you tried on a small board. Then take exactly four pennies and find a way to place them below the line so that a penny lands on the star. Draw every army of four that works on a small board, with its jumps drawn as arrows numbered 1, 2, 3.

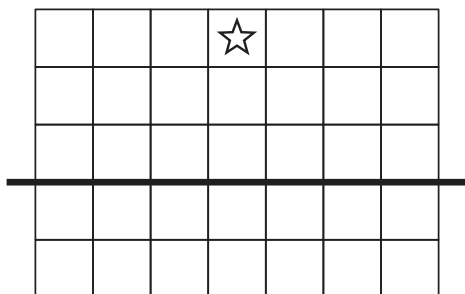
P

Problem 9: On board Q the star is three rows above the line. Put a penny on each of the eight circles. Find seven jumps that land a penny on the star. Record them in order: draw the first jump as an arrow on small board 1, the second on small board 2, and so on to small board 7.

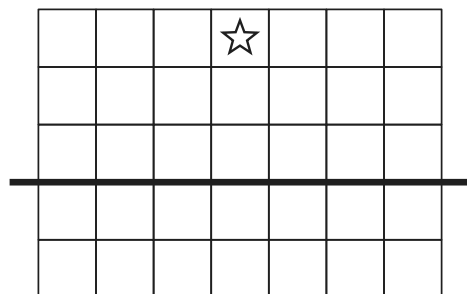
Q



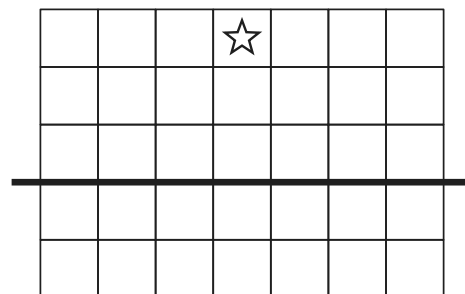
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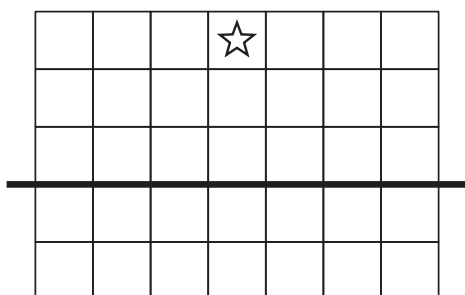
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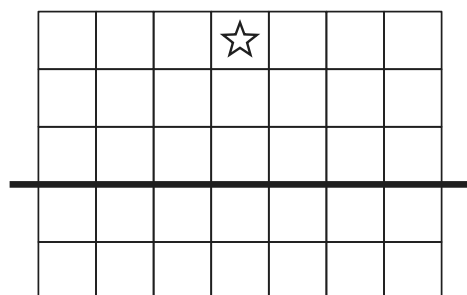
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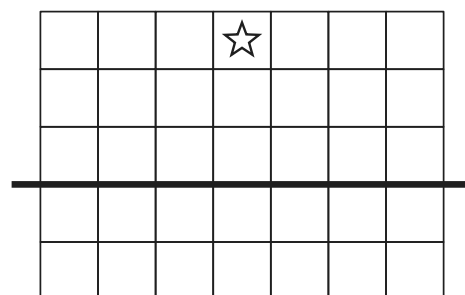
4



5



6



7

